

Motivation & Problem

Pneumonia is a leading cause of respiratory mortality. Manual X-ray interpretation is slow and expert-dependent. Most AI solutions classify without explanation and are never deployed.

Objectives

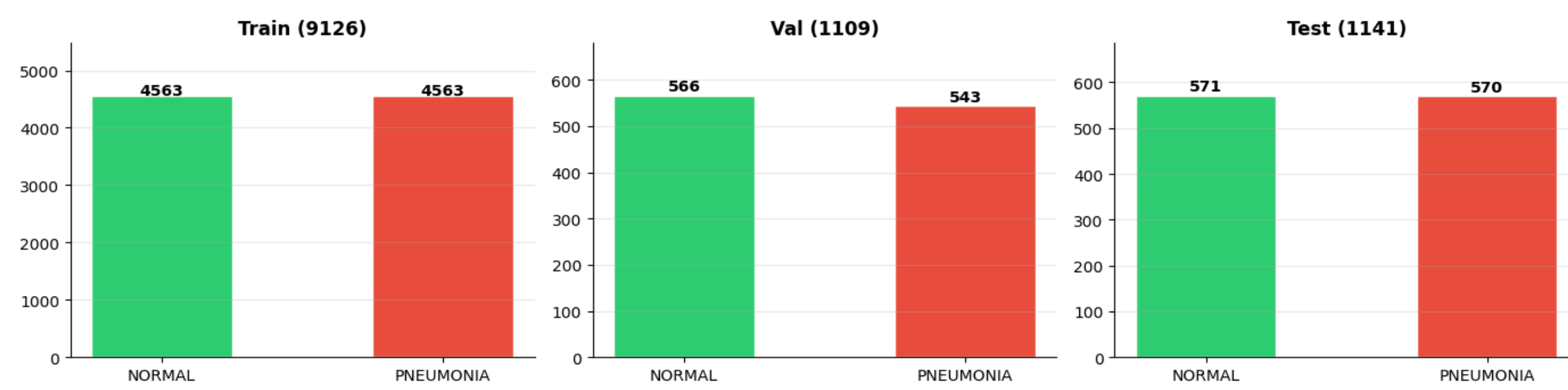
- Binary classifier: NORMAL vs PNEUMONIA
- Explain predictions with [Grad-CAM heatmaps](#)
- Deploy via [FastAPI](#) REST backend
- Deliver results through an interactive [web app](#)

Deep Learning

Explainable AI

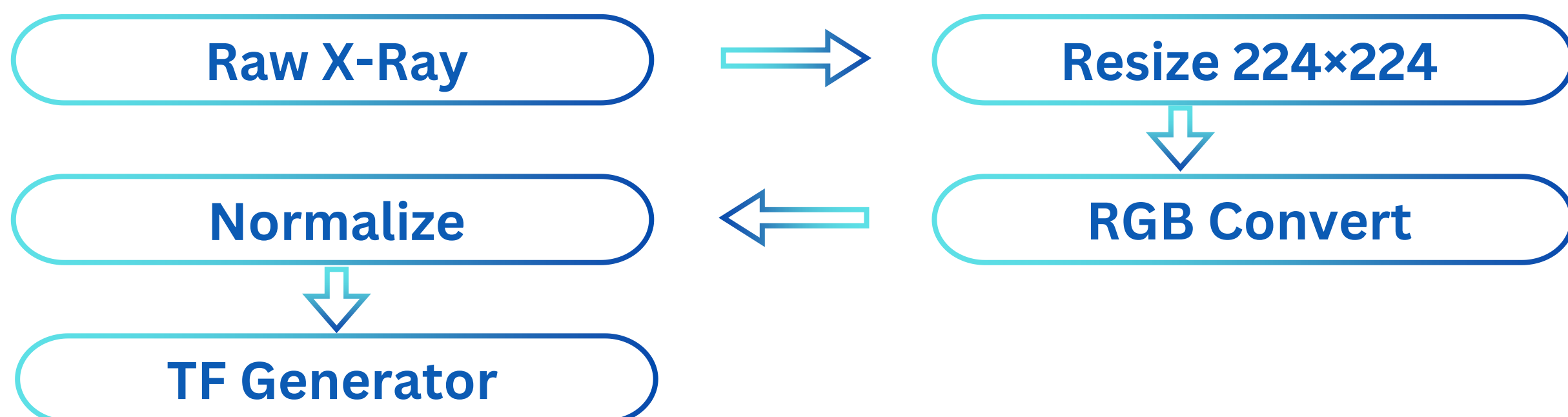
Medical Imaging

Dataset & Preprocessing



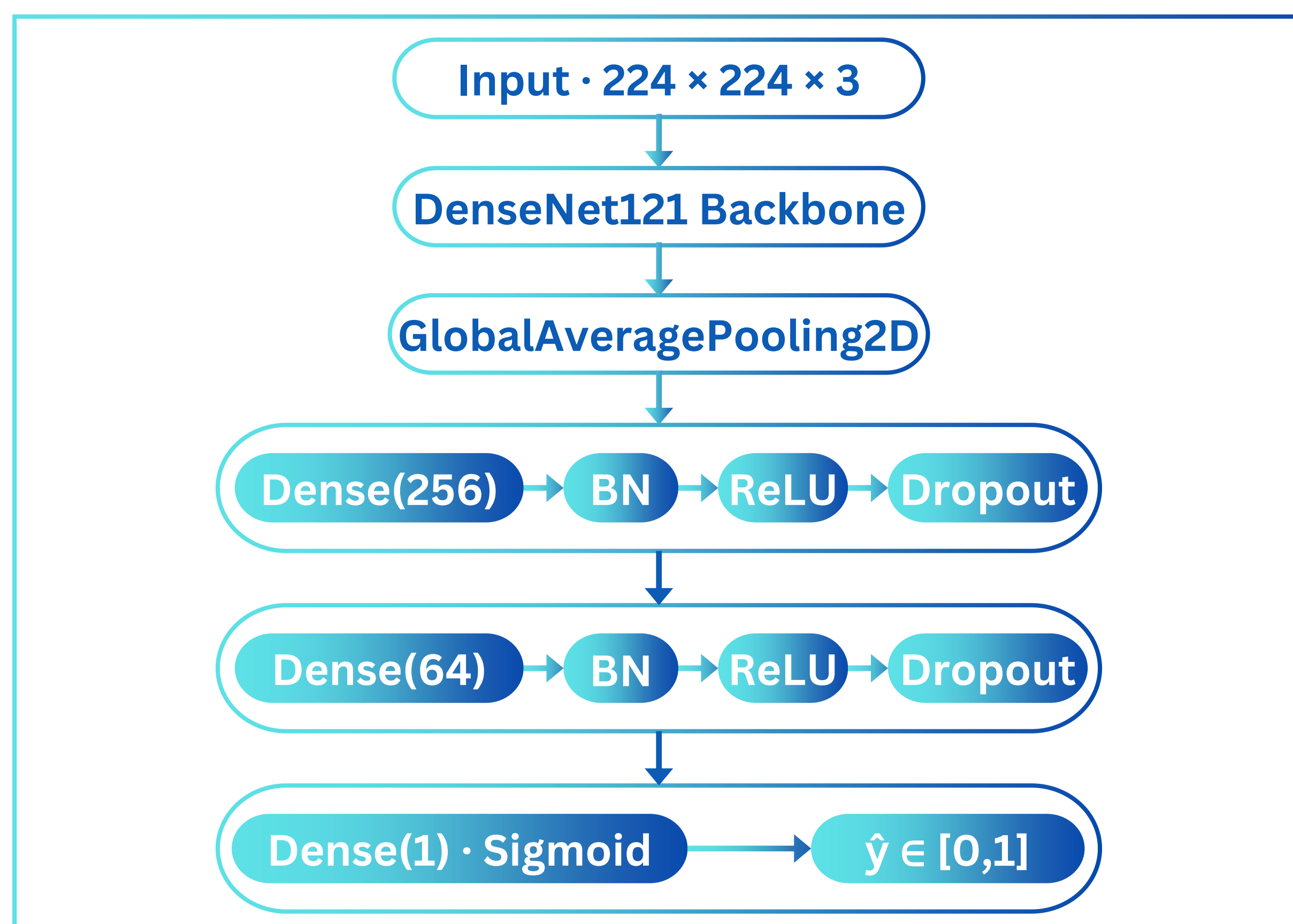
Balanced Chest ray - Binary labels - Balanced classes

Preprocessing Pipeline



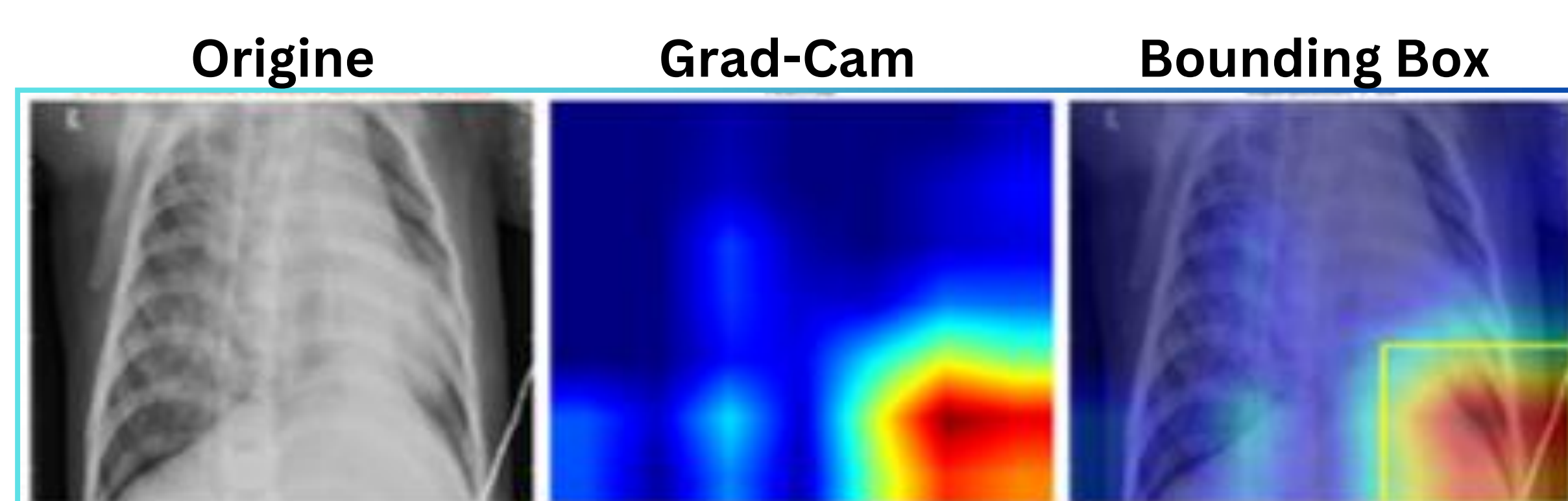
Methodology

Model Architecture (DenseNet121)



- **Two-phase training:**
 - Transfer Learning (frozen backbone)
 - Fine-Tuning (last 50 layers unfrozen)
- **Hyperparameter** optimization via Grid Search
- **EarlyStopping + ReduceLROnPlateau**
- Optimal decision threshold selected using F1-score maximization

Explainable AI — Grad-CAM



Grad-CAM visualizes which lung regions drove the prediction critical for clinical trust. Target: last Conv2D of DenseNet121.

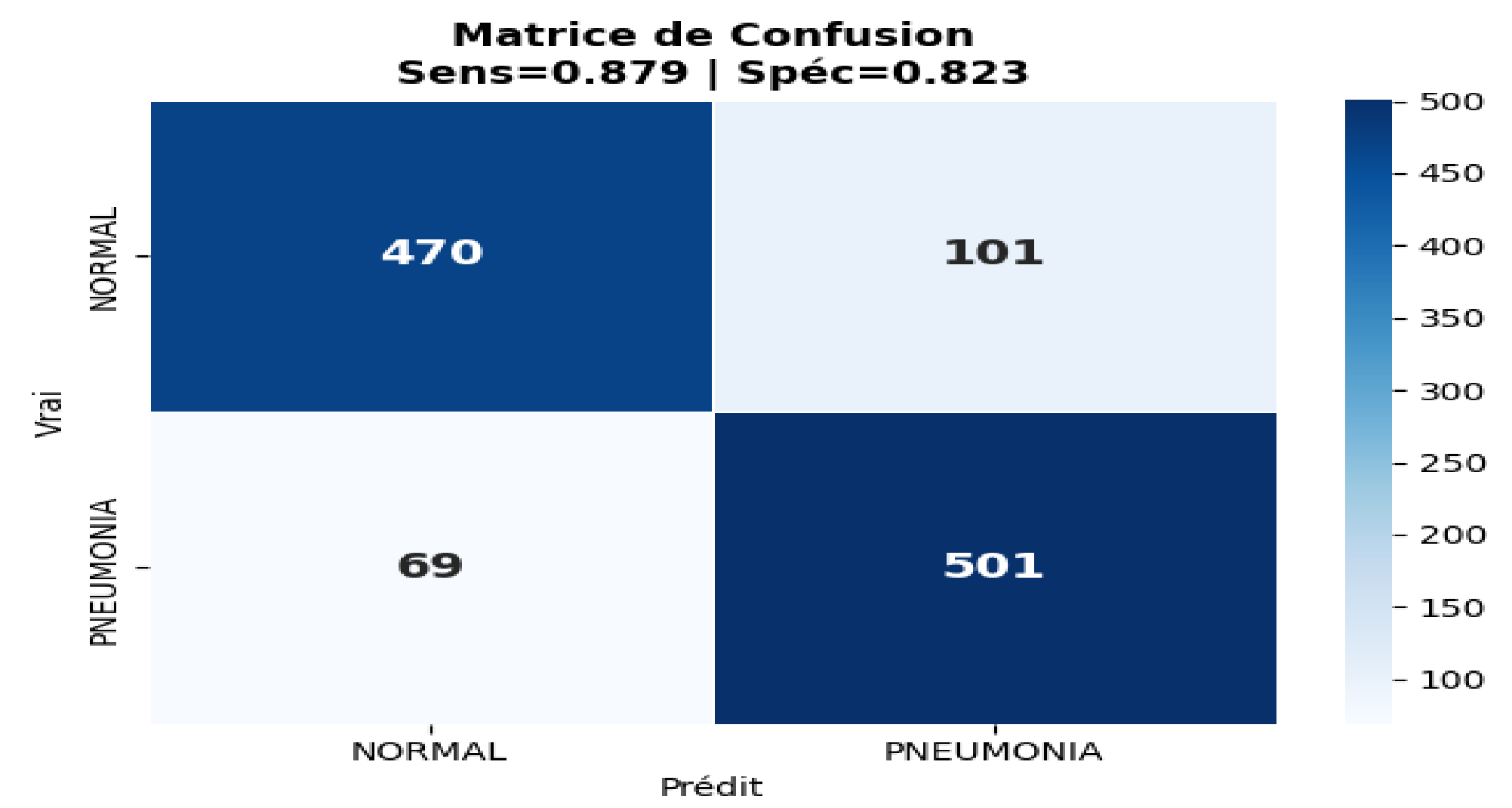
Test Set Metrics

85.10 %
Accuracy

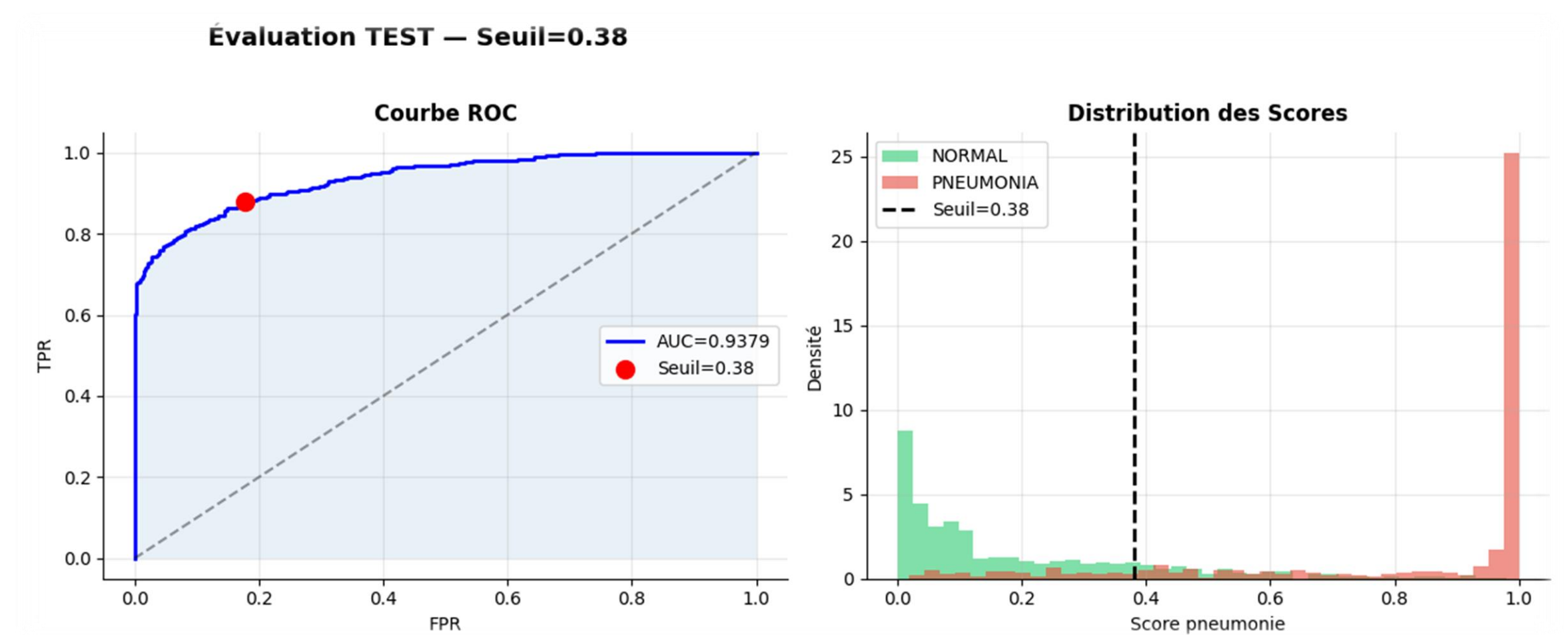
93.79 %
AUC-ROC

85.49 %
F1-Score

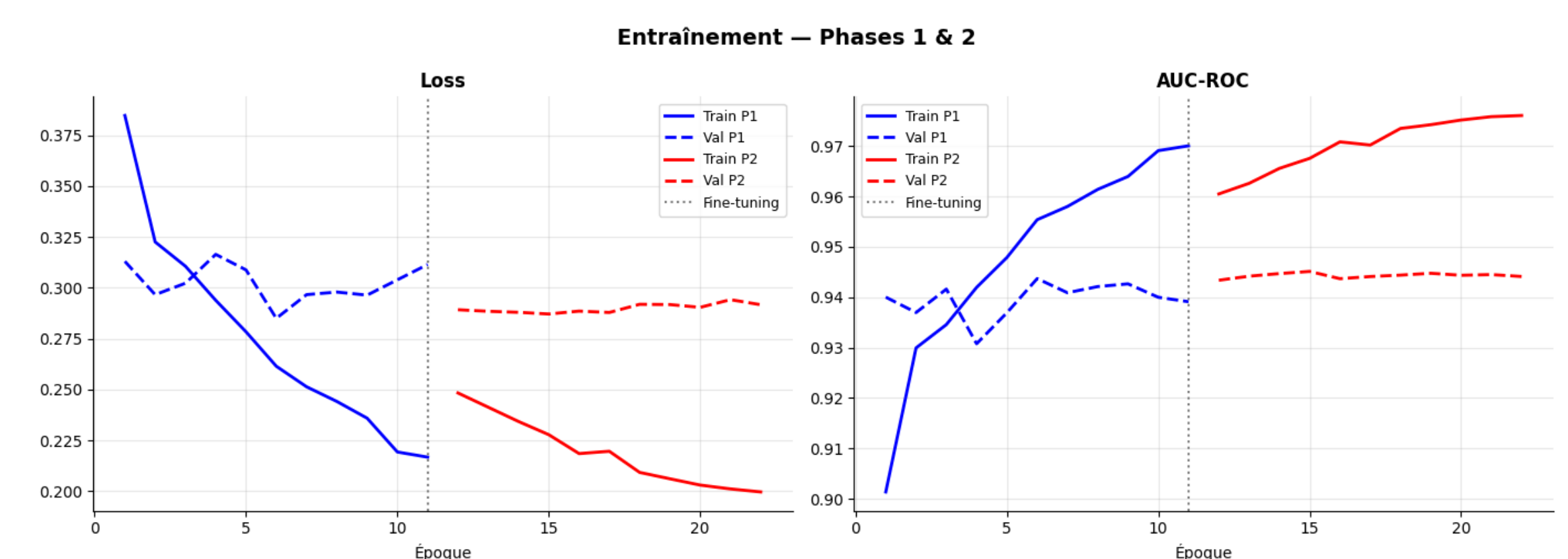
Confusion Matrix



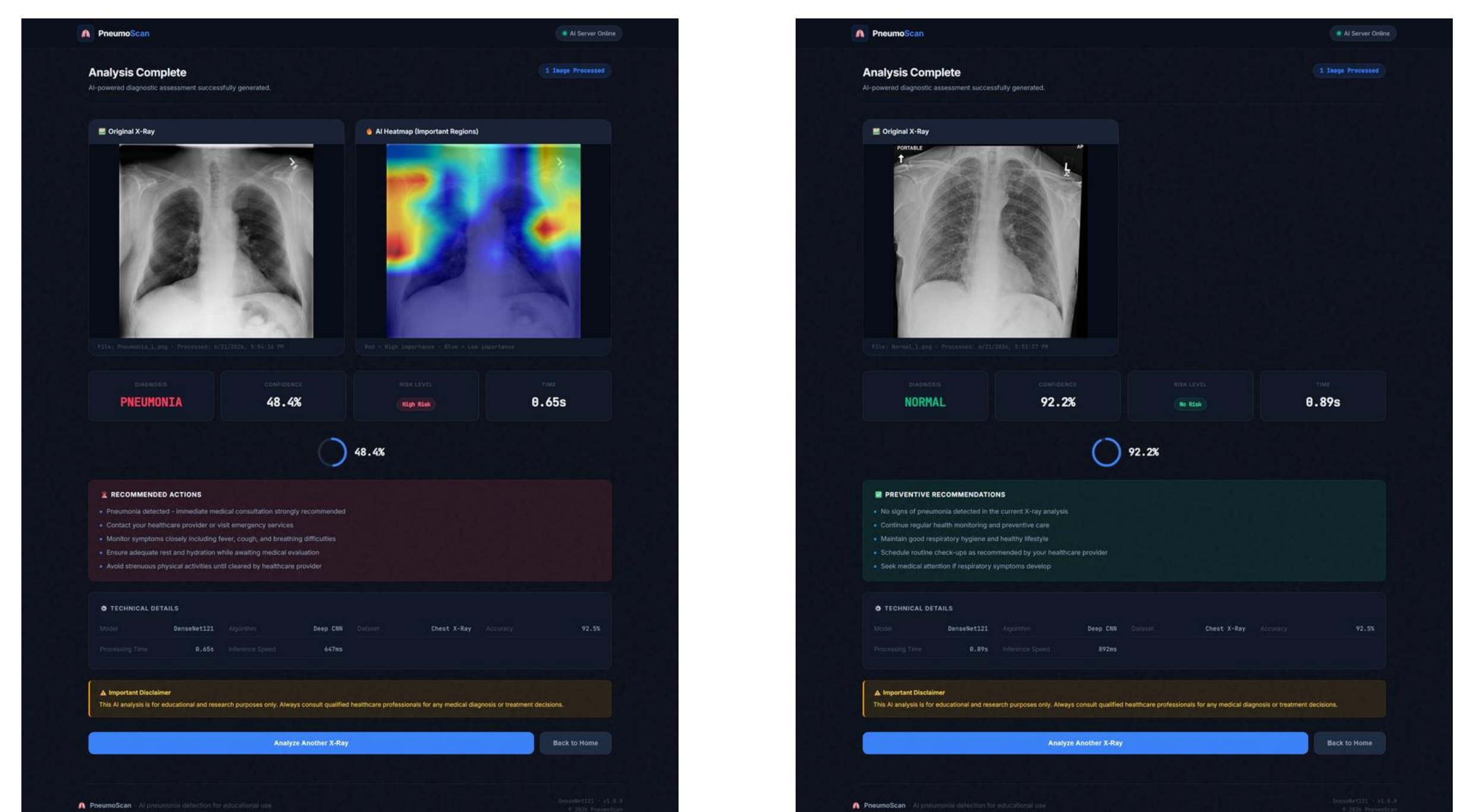
ROC & Score



Training History



WEB APP



Conclusion

In conclusion, a DenseNet121 model achieves state-of-the-art accuracy on a balanced Kaggle dataset, with Grad-CAM-based localization of lung regions, deployed via FastAPI and a web interface, and rigorously evaluated using ROC, PR curves, confusion matrix, grid search, and F1-threshold optimization.



SCAN ME



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